

Experiment 1 — Does repository context improve LLM code review?

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Question

Does augmenting an LLM code reviewer with a repository **knowledge graph** (KG) and/or **retrieval-augmented generation** (RAG) produce better PR review comments than the prevailing diff-only baseline?

Design

Dataset	40 real merged pull requests from 5 open-source repositories (Apache Kafka, Godot, Grafana, scikit-learn, Jenkins)
Modes compared	baseline (diff-only) · kg · rag · hybrid (kg+rag)
Generation	gpt-4o, T = 0.4, seed = 42 — identical prompts except the injected context
Scoring	3-judge LLM panel (gpt-4o-mini, gpt-4o, gemini-2.5-flash), majority vote per criterion
Rubric	25 criteria grounded in Bacchelli & Bird (ICSE’13), Bosu et al. (MSR’15), Sadowski et al. (ICSE-SEIP’18), ISO/IEC 25010; a pre-defined 9-criterion KG-relevant subset (impact/dependency criteria the KG is designed to help with)
Statistics	percentile bootstrap (B = 10 000) for CIs; paired sign-flip permutation tests (B = 20 000), seed 2026

Results

Per-mode means (95% bootstrap CI), n = 40:

Mode	Total /25	KG-relevant /9
baseline	9.20 [8.70, 9.70]	4.97 [4.67, 5.28]
kg	9.82 [9.28, 10.43]	5.58 [5.22, 5.95]
rag	10.07 [9.57, 10.60]	5.40 [5.00, 5.78]
hybrid	9.93 [9.47, 10.38]	5.50 [5.22, 5.78]

Paired differences vs baseline (same PR):

Mode	Total Δ	p	d_z	KG-relevant Δ	p	d_z
kg	+0.62 [+0.05, +1.20]	0.054	0.33	+0.60 [+0.23, +0.97]	0.007	0.47
rag	+0.88 [+0.30, +1.45]	0.007	0.47	+0.42 [+0.05, +0.82]	0.057	0.33
hybrid	+0.72 [+0.10, +1.35]	0.037	0.36	+0.53 [+0.17, +0.90]	0.008	0.46

Inter-judge agreement is substantial: Cohen's $\kappa = 0.59$ – 0.72 across the three judge pairs.

Reading

Each context type improves the metric it targets, and the hybrid improves both. KG's gain concentrates on the KG-relevant criteria (dependency/impact grounding, +0.60 of 9, $p = .007$, medium effect); RAG's gain is on total coverage (+0.88 of 25, $p = .007$). Neither dominates the other — the defensible claim is **complementarity**, not “graphs beat retrieval”.

Known limitations → what Experiment 2 addresses

1. The result is **correlational**: real PRs vary in a hundred ways, so rubric gains do not prove the KG causes better *defect-relevant* reviews.

2. Scores are **LLM-judged** against a rubric, not against ground truth.

Experiment 2 (separate document) closes both gaps with a controlled defect-injection design: known planted defects, objective ground truth, and judge validation against a deterministic oracle. A small human study ([link in the accompanying message](#)) then closes the remaining machines-grading-machines loop.